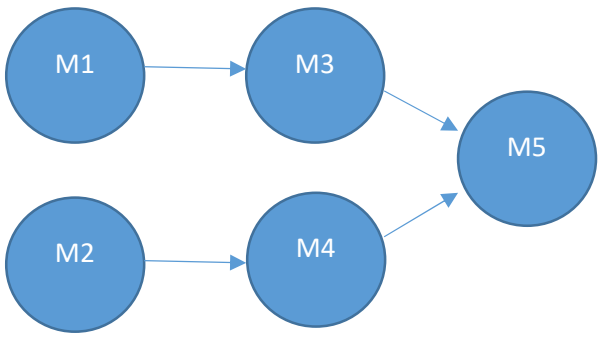
Exercise 1

Define an algorithm in Matlab for the following scheduling problem.

Schedule the following *n* jobs, where p_j is the processing time on machine i. The goal is to minimize Cmax.

Job	J1	J2	J3	J4	J5	J6	J7	J8	J9	J10
P_1	5	3	6	8	4	12	12	5	3	2
P_2	12	6	1	5	6	15	3	2	8	8
P_3	1	20	2	5	7	11	12	2	5	4
P_4	13	10	1	15	6	12	11	4	4	13
P_5	2	6	2	1	5	13	2	7	18	3



Verify that the solution that has been obtained is optimal or not comparing with the solution obtained in a mathematical programming problem defined in Excel (or other spreadsheet tool with optimization module) or Lingo or Cplex or Matlab.

Organise the input/output information in a SQL database. Implement also a procedure to initialize new random instances of the problem on n jobs